HUSEYIN DEMIR GOKSU

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EDUCATION

Harvard-Westlake School, Los Angeles, CA *Class of 2027*

- Honor Roll (2023 - 2026), GPA: 4.23 weighted / 3.94 unweighted | SAT: 1540
- **Relevant Coursework:** Calculus I & II (BC-level), Honors Physics C: Electricity & Magnetism, Honors Topics in Computer Science, Honors Economics, Robotics Varsity–FRC, Honors Precalculus, Honors Physics I, Advanced Computer Science, Honors Design and Data Structures. **Directed Study:** Topics in Modern Physics.
- **AP Examinations (independently prepared, AP Scholar):** Computer Science A (5), Physics 1 (5), Precalculus (5), Calculus BC (IP), Physics C: Electricity & Magnetism (IP), Microeconomics (IP), Macroeconomics (IP)

UCLA Engineering Design Summer Institute: Aerospace Engineering (Rockets) *Jun 2024*

Engineering 1RK, 2.0 UCLA units; 3-week precollege program

- Designed and simulated rockets in OpenRocket and CAD; built and launched multiple 3D-printed rockets, and one fiberglass high-powered rocket.
- Collaborated on a team high-powered rocket from design through launch, reaching ~2800 ft.

MIT Beaver Works Summer Institute (BWSI) Autonomous Air Vehicle Racing (Online Course) *Jan – Feb 2025*

- Completed online curriculum covering mathematics and Python programming behind autonomous drone (quadrotor) control algorithms.

ENGINEERING RESEARCH & INDEPENDENT PROJECTS

Independent Research, Anisotropy-Aware Topology Optimization for 3D-Printed Parts *Jul 2025 – Present*

Polygence Research Program, mentored by Kyle Buznitsky, Ph.D. in Mechanical Engineering, MIT.

- Extended an open-source topology optimization code (PyTopo3D) to account for the direction-dependent (anisotropic) material properties of FDM 3D-printed parts.
- Designed and built a custom aluminum video extensometer tensile testing rig with camera calibration and marker tracking.
- Conducted 50+ tensile tests on aluminum and PLA specimens, producing 30 valid data sets after quality screening.
- Benchmarked the rig against aluminum and measured PLA properties in three print orientations, finding stronger anisotropy than published values.
- Currently testing topology-optimized PLA parts; preparing results for publication.

Independent Project, Robotic Arm *2022 – 2024*

- Independently designed, 3D-printed, and assembled a motorized robotic arm over 18 months, over three iterations.
- Used six stepper motors with planetary gearboxes and belt drives, an Arduino microcontroller, and a custom PCB for motor control.

INDIVIDUAL HONORS & DISTINCTIONS

FIRST Leadership Award Semifinalist (formerly FIRST Dean's List) *Mar 2026*

- Selected by judges at the CA District Glendale Event and advanced to a finalist interview at the FIRST California Southern State Championship for FIRST Robotics Competition's highest individual student honor.

National Merit Scholarship Semifinalist, Class of 2027 *Oct 2026*

Team FIRST Award, FRC Team 1148 *May 2026*

- Awarded by the team for building team culture and mentoring students inside and outside the school robotics community.

Skill Grinder Award, FRC Team 1148 *May 2025*

- Awarded by the team for outstanding design contributions and technical skill.

AP Scholar Award, College Board *Jul 2025*

- Awarded for a score of 3 or higher for three AP exams.

COMPETITIVE ROBOTICS & ENGINEERING LEADERSHIP

FIRST Robotics Competition (FRC), Varsity Robotics Team 1148, Harvard-Westlake School *Sept 2024 – Present*

***Senior Advisor**, 15 hours/week in season, 8 hours/week off-season* *Sept 2026 – Jun 2027*

- Proposed and established the Senior Advisor role to support leadership transitions.Provided strategic engineering guidance to a 10-member design team, advising on robot design, technical troubleshooting, and integration across mechanical, electrical, programming, and manufacturing subteams.

- Mentored approximately 20 rookie members through hands-on engineering workshops, developing CAD proficiency and foundational design skills.
- Advised the Design Subteam Lead on technical decisions and troubleshooting, helping transfer leadership responsibilities to the next generation of student engineers.
- Documented engineering workflows, design standards, and lessons learned from previous seasons to preserve institutional knowledge and ensure continuity across leadership transitions.

Design Lead & Technician**,** *20+ hours/week* *Sept 2025 – Jun 2026*

- Led a 6-member engineering design team, overseeing robot development, CAD workflows, and technical decision-making.
- Translated competition objectives into engineering requirements, coordinated subsystem integration across teams, and led the creation of the robot's technical binder.
- Designed, prototyped, and assembled robot subsystems; documented design iterations and collaborated with other subteams to resolve engineering challenges.
- Developed and delivered engineering design training to 15 students, strengthening CAD proficiency and foundational mechanical design skills.
- Served as Pit Technician at four competitions, diagnosing and repairing mechanical failures under time pressure to maintain the robot's on-field performance.

Design Team Member, Pit Crew, Lead Scouter, *20 hours/week* *Sept 2024 – Jun 2025*

- Led the design of the complete off-season robot using Onshape CAD, integrating mechanical subsystems into a cohesive system.
- Designed and prototyped the competition robot's intake and scoring mechanisms, contributing to game-piece acquisition and scoring functionality.
- Collaborated with the manufacturing subteam to translate CAD designs into fabrication plans, identify procurement needs, and support robot assembly.
- Conducted technical scouting and robot performance analysis, evaluating mechanical designs, scoring capabilities, and system reliability to inform match strategy and alliance selection.

VEX Robotics Competition, Team 62A, Harvard-Westlake School *Sept 2023 – Jun 2024*

Build Lead, *20 hours/week*

- Led the design and construction of three competition robots, overseeing prototyping, testing, and iterative improvements to enhance performance.

FIRST Tech Challenge, Team 19299, Beverly Vista Middle School *Sept 2022 – May 2023*

Team Captain, *9 hours/week*

- Led the design, assembly, and testing of the competition robot, troubleshooting mechanical issues and performing on-site repairs during competitions to maintain robot performance.

TEAM AWARDS & COMPETITION ACHIEVEMENTS

FIRST Robotics Competition – Team 1148 *2024 – Present*

- Judges' Award, FIRST California Southern State Championship
- Excellence in Engineering Award, CA District Glendale Event
- Regional Event Finalist, Hueneme Port Event
- Offseason Event Finalist, Tidal Tumble
- Competed at the FIRST Championship (World Championship), Curie Division, Houston
- Engineering Inspiration Award, Idaho Regional; qualified the team for the FIRST Championship
- Innovation in Control Award, Los Angeles Regional

VEX Robotics Competition – Team 62A *2023 – 2024*

- Competed in 13 tournaments, including two Signature Events, the California Regional Championship, and the VEX Robotics World Championship (Technology Division, Round of 16).
- Earned eight team awards and distinctions, including Tournament Champion and Finalist titles, Excellence, Design, Innovate, and Judges Awards.

LEADERSHIP, MENTORSHIP & COMMUNITY IMPACT

Mentor, FRC Team 12018 "Avartre," Hatay, Türkiye *Aug 2026 – Present*

- Began supporting team Avartre, a newly forming FRC team in Hatay: shared Turkish-language FRC documentation and self-produced Onshape training videos.

- Advised on grant-seeking strategies.

Design Mentor, FRC Team 10947 "Antioch Rising," Hatay Science High School, Türkiye *Nov 2025 – Present*

- Developed and taught a 10-module Turkish-language curriculum covering Onshape CAD, FRC components, and engineering fundamentals for a robotics team in earthquake-affected Hatay, Türkiye; Uploaded 10 Turkish Onshape and Design Education meetings publicly on YouTube.
- Mentor the team's five-member design subteam through weekly 1.5-hour virtual sessions, guiding students toward independently designing their 2027 competition robot.
- Supported the team's development during its rookie season, when it earned the Team Spirit Award at the Marmara Regional despite lacking an in-person technical mentor.
- Organized a virtual meeting with Carnegie Mellon University Professor Zeynep Temel in May 2026.

Co-President, Member, Rocketry Club, Harvard-Westlake School *Oct 2024 – Present*

- **As Co-President (2025 – present):** Leading the design, construction of a Level 2 high-power certification rocket, overseeing aerodynamics and mechanical engineering, launch planned May 2027
- Founded rocketry workshop outreach event for middle school students, developing an age-appropriate rocket science curriculum and designing custom 3D-printable rocket kits.
- 12 hours of hands-on workshops for 26 students at HOLA (14) and the Bresee Foundation (12), guiding students through assembling and launching their own rockets.
- **As Member (2024 – 2025):** Researched rocket designs and developed an original design proposal for the club's consideration.

Co-President, Founding Member, Turkish Club, Harvard-Westlake School *Oct 2024 – Present*

- Co-founded the club with four Turkish students; promoted the club and recruited members to raise awareness of Turkish culture, growing attendance from 7–8 to 20 students over two years.
- Organized a virtual speaker series featuring Turkish-American scientists Prof. Elza Erkip (NYU), Prof. Tamer Başar (UIUC), and Prof. Betül Kaçar (UW–Madison); identified and invited speakers, researched their work, and prepared discussion questions.
- Lead discussions and cultural activities exploring Turkish history, politics, geography, language, and arts.
- Independently launched a social media fundraising campaign, raising **$3,500** for the Aziz & Gwen Sancar Foundation's Sancar Scholars Residence.

Los Angeles Coordinator, Team Leader & Conversation Partner, Turks Without Borders *Jul 2026 – Present*

- Launched the TWB English conversation program in California by establishing a partnership with the TWB central team in Türkiye.
- Built connections within the Los Angeles Turkish-American community to publicize the program and promoted it through the school's Turkish Club.
- Recruited 20 Turkish-American students from Los Angeles and surrounding areas and matched them with 20 students from five schools across Türkiye.
- Serve as team leader for the Hatay Science High School group while volunteering weekly as a conversation partner, facilitating English practice and cultural exchange.

VEX Robotics Referee, Harvard-Westlake Challenge, Harvard-Westlake School *Dec 2024, Dec 2025*

- Served as a referee for two consecutive years at Harvard-Westlake's two-day VEX Robotics tournaments, officiating matches for Southern California teams competing to qualify for the State Championship.

Paid Summer Camp Coach, VEX Robotics Summer Camp, Harvard-Westlake School *Jun 2025*

- Taught robotics fundamentals (design, building, and programming) to 30 beginner students in a 2-week summer camp.
- Guided students as they built their own robots and competed with them in the final tournament.

SKILLS, LANGUAGES & INTERESTS

Programming: Python, Java
Engineering Tools: Onshape, Fusion 360, SolidWorks (CAD); ANSYS Mechanical (finite element analysis); OpenRocket (rocket design and simulation); KiCad (PCB design); 3D printing and rapid prototyping
Visual Arts: Artwork selected for the school's annual student art exhibition each year; completed Drawing and Painting I and II (grades 10–11); enrolled in Honors Senior Seminar in Drawing (grade 12).
Languages: English (native-level), Turkish (native), Spanish (intermediate)
Interests: Golf, drawing, cooking.